The Fricke involution on the 11a1 model is P ↦ T − P

On the genus-one curve X_0(11), using the model

E: $y^2 + y = x^3 - x^2 - 10x - 20$, with cusp infinity as the group origin, the moduli-theoretic Fricke involution is the global algebraic map

W_11(P) = T − P, where T=(16:-61:1) is cusp zero.

This identifies the candidate coordinate map used by the D=-2631 pointwise audit with the modular involution throughout the curve, and hence after good reduction in every characteristic other than 11. It turns that audit's finite-class commutation check into evidence about the actual Fricke action on the saved Hecke fibers.

Proof

The explicit modular-curve model and cusp table, Section 5.1, give E, cusp infinity O, and cusp zero T=(16,-61) in these coordinates. The moduli definition of the Atkin–Lehner involution dualizes a cyclic 11-isogeny, so W_11(O)=T and W_11(T)=O.

Any automorphism w of an elliptic curve is P ↦ w(O)+u(P) for an origin-preserving automorphism u. The j-invariant of this curve is −122023936/161051, different from 0 and 1728; in characteristic zero its origin-preserving automorphisms are +1 and −1. Sage verifies T has order five. If u=+1, then w(T)=2T, contradicting the cusp exchange w(T)=O. Therefore u=−1, and W_11(P)=T−P for every geometric point.

The unique normalized weight-two newform at level 11 has Fricke eigenvalue −1, independently agreeing with the action on invariant differentials; the LMFDB newform table records this sign. Sage also checks the modular degree of the 11a1 parametrization is one, so these exact coordinates give the curve itself.

Reproducibility

The frozen protocol fixes the model and theorem before the checks. Run ./sage --runtime-info and ./sage -python experiments/hecke-x011-global-fricke-20261026/verify.py from the repository root. The saved runtime and verification show conductor 11, modular degree one, T of order five, Fricke sign −1, and a_11=1. The map is an involution on every checked point over F_7, F_13, and F_23; the F_23743 check uses three distinguished points. The Q-level proof above supplies the global identity; these finite-field calculations check the implementation.

The Lean module proves the decisive conditional algebraic step: an affine map with linear part ±1 that exchanges zero and an element of order greater than two has linear part −1. It also proves the resulting map is an involution and exchanges the cusps. lake env lean FrickeAffine.lean passed under Lean v4.35.0-rc4, pinned Mathlib revision 021c46bdd042f67b538b9f9871d556a93713a9a9; #print axioms reports only propext for all four declarations. lean4checker --fresh was unavailable. The full modular theorem's Lean bridge is still open: formal definitions of X_0(11), its Fricke moduli automorphism, this Weierstrass coordinate isomorphism, and the classification of origin-preserving automorphisms for the curve are not in this module. The written proof uses those mathematical inputs explicitly.

Visual report and graph review (2026-10-30)

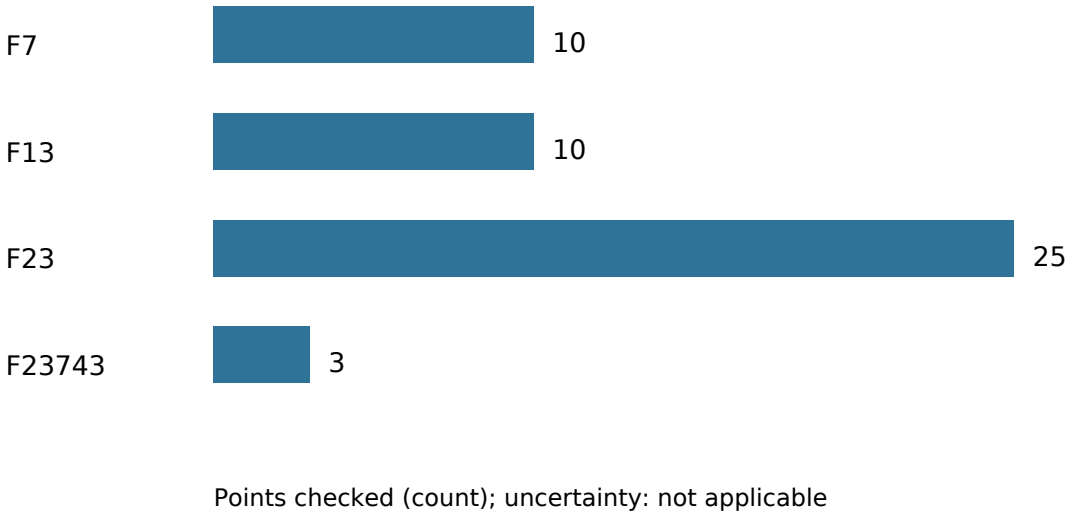
The PDF report contains this report, the global-identity evidence diagram (editable Graphviz source), and the saved finite-field coverage plot. Rebuild it with python3 experiments/hecke-x011-global-fricke-20261026/build_report.py; the renderer requires Graphviz, ReportLab and PyMuPDF. The rendering receipt records versions, exact source hashes and the saved coverage counts. This documentation build does not rerun Sage or Lean and supplies no new experimental or proof evidence.

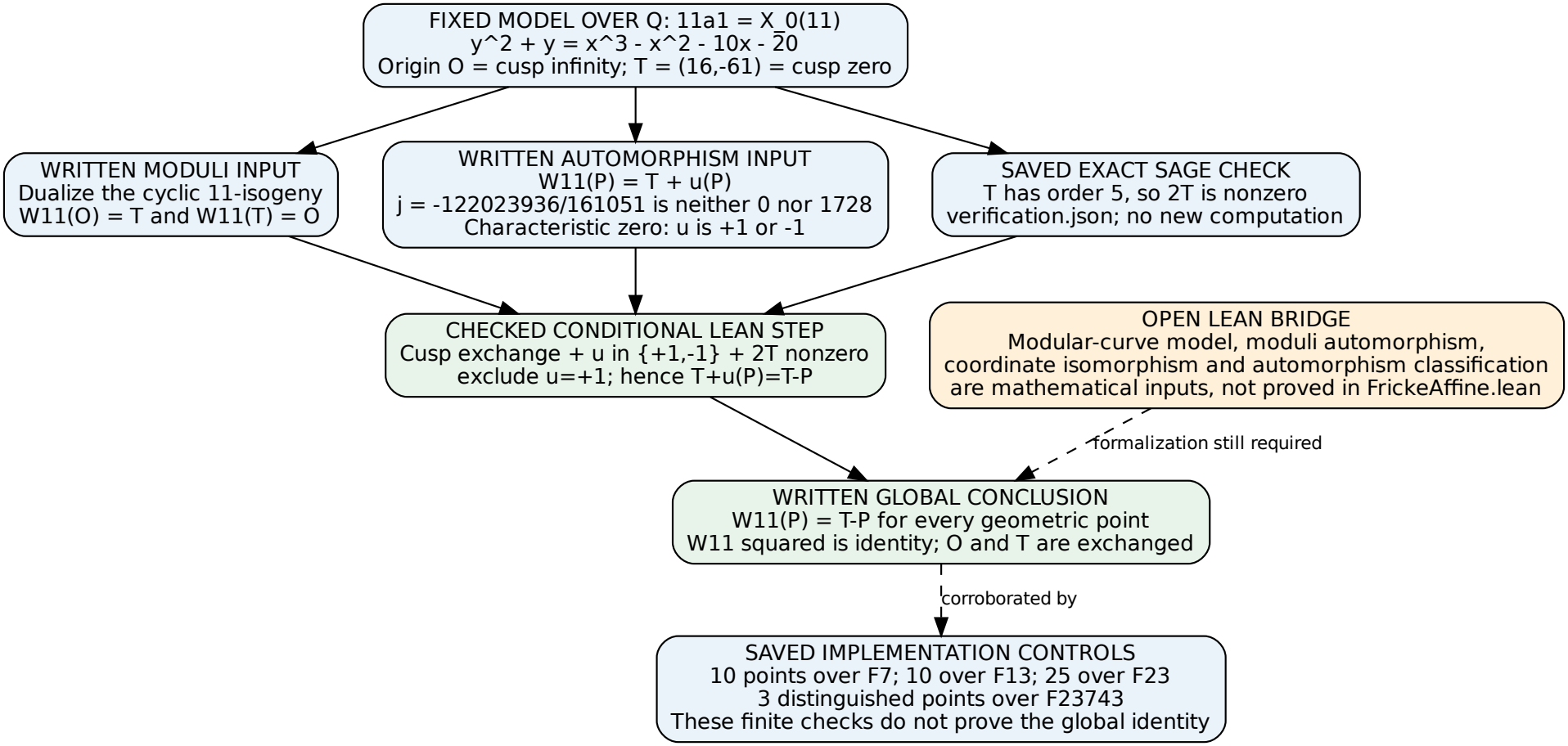
The diagram distinguishes the written modular-curve inputs, the conditional Lean step and the remaining formalization bridge. The plot reports exact saved point counts, with no statistical uncertainty: 10, 10 and 25 enumerated points over F7, F13 and F23, and only three distinguished points over F23743. Those checks corroborate the implementation; they do not prove the global identity or establish an ECDLP speedup.

Affected-graph check: inspected the sources for experiments/volcano-ic/figures/census.svg and ecdlp.svg (Koblitz descendant counts, yield and solve-work comparisons), and experiments/hecke-d39-scalar-closure-20261030/evidence.dot and result.dot (the fixed F367 rank/control fixture). Their source data and finite claims do not change under this documentation correction, so their figures are retained. The associated experiments/ic-candidate-catalog/curves.yaml was checked; this addition creates no curve record, isogeny route, subgroup-transport claim or ISO1 admission. The new diagram records the global Fricke interpretation without relabeling a conditional Lean lemma as the complete modular theorem.

Saved finite-field coverage

Source: verification.json. Counts are exact saved control coverage, not timing measurements or a proof of the global modular identity. The F23743 check uses three distinguished points; the other fields enumerate every curve point.




FIXED MODEL OVER Q: 11a1 = X_0(11)
y^2 + y = x^3 - x^2 - 10x - 20
Origin O = cusp infinity; T = (16,-61) = cusp zero
WRITTEN MODULI INPUT
Dualize the cyclic 11-isogeny
W11(O) = T and W11(T) = O
WRITTEN AUTOMORPHISM INPUT
W11(P) = T + u(P)
j = -122023936/161051 is neither 0 nor 1728
Characteristic zero: u is +1 or -1
SAVED EXACT SAGE CHECK
T has order 5, so 2T is nonzero
verification.json; no new computation
CHECKED CONDITIONAL LEAN STEP
Cusp exchange + u in {+1,-1} + 2T nonzero
exclude u=+1; hence T+u(P)=T-P
OPEN LEAN BRIDGE
Modular-curve model, moduli automorphism,
coordinate isomorphism and automorphism classification
are mathematical inputs, not proved in FrickeAffine.lean
formalization still required
WRITTEN GLOBAL CONCLUSION
W11(P) = T-P for every geometric point
W11 squared is identity; O and T are exchanged
corroborated by
SAVED IMPLEMENTATION CONTROLS
10 points over F7; 10 over F13; 25 over F23
3 distinguished points over F23743
These finite checks do not prove the global identity